Tensor computations

Nonnegative H-eigenvalue inheritance from odd-order Hankel tensors

Difficulty: challenging **Importance:** interesting to specialist

Statement

For every odd $m \geq 3$, integer $q \geq 2$, dimension $n \geq 2$, and vector $h \in \mathbb{R}^{qm(n-1)+1}$, define Hankel tensors A and B with this same generating vector. Their orders and dimensions are

$$\text{order}(A) = m, \quad \dim(A) = q(n-1)+1, \qquad \text{order}(B) = qm, \quad \dim(B) = n.$$

An order-s, dimension-N Hankel tensor has entry $h_{i_1+\cdots+i_s-s}$, with each index in $\{1, \ldots, N\}$.

For a real order-s tensor C, an H-eigenvalue is a real λ admitting a real nonzero vector x such that, for every i,

$$\sum_{i_2,\ldots,i_s=1}^{N} C_{ii_2\cdots i_s} x_{i_2} \cdots x_{i_s} = \lambda x_i^{s-1}.$$

Conjecture: if A has no negative H-eigenvalues, then B has no negative H-eigenvalues.

Relevance

The question transfers a spectral positivity certificate between different Hankel representations of the same data, relevant to structured tensor eigenvalue computation and polynomial positivity.

References

1. W. Ding, L. Qi, and Y. Wei, *Inheritance properties and sum-of-squares decomposition of Hankel tensors: theory and algorithms*, BIT Numer. Math. 57 (2017), 169–190. DOI; author PDF, final §4, "The third inheritance property of Hankel tensors," concluding conjecture; §2 gives order/dimension conventions.
2. L. Qi, *Hankel Tensors: Associated Hankel Matrices and Vandermonde Decomposition*, 2014. Primary preprint, §5, concerning H-eigenvalues and complete Hankel tensors.

Status check — 2026-09-08

Reference 1 proves the corresponding inheritance with even lower order and leaves the odd-order case conjectural. Searches *"Hankel" "third inheritance" counterexample*, *"tensor" "inheritance" "conjecture" Hankel proof*, and *"Hankel" "inheritance" "2026" conjecture* located no resolution. Results under stronger assumptions such as complete Hankel structure do not establish this statement. This is a bounded historical-source check.